

PROJECT DOCUMENTATION

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

ServiceNow Project

Team ID: SWTID-2026-2467

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1. Introduction

Project Title: Auto Ticket Classification using Flow Designer

Team Members:

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2. Project Overview

Purpose: The project automates the classification of common IT support tickets using ServiceNow Flow Designer. It is designed to reduce repetitive manual classification, improve consistency and maintain structured ticket records.

Features:

- Incident Workflow ticket creation and structured ticket fields.
- Automatic classification using Flow Designer conditional logic.
- Configured classification for Wi-Fi, Projector, Forgot Password and Slow Computer issues.
- Category and Subcategory configuration with dependent-field control.
- Automatic Update Record actions.
- Ticket confirmation notification.
- Access control configuration for the custom table.
- Project Update Set for capturing configuration changes.

3. Architecture

The project architecture is based on native ServiceNow components rather than a separate frontend/backend/database application.

- Application Layer: Incident Workflow form and custom Incident Workflow table.
- Automation Layer: ServiceNow Flow Designer containing the trigger, conditional branches and Update Record actions.
- Data/Configuration Layer: Category, Subcategory and dependency configuration.
- Security/Deployment Layer: Access Controls and Project Update Set.

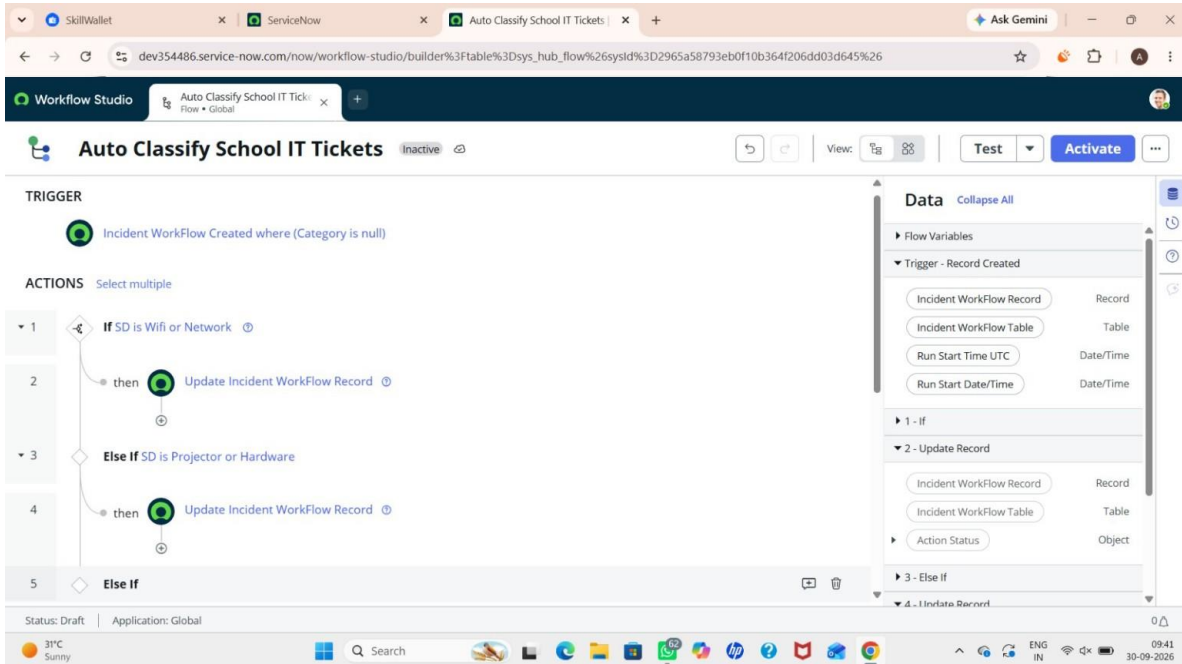


Figure 1. Flow Designer classification automation.

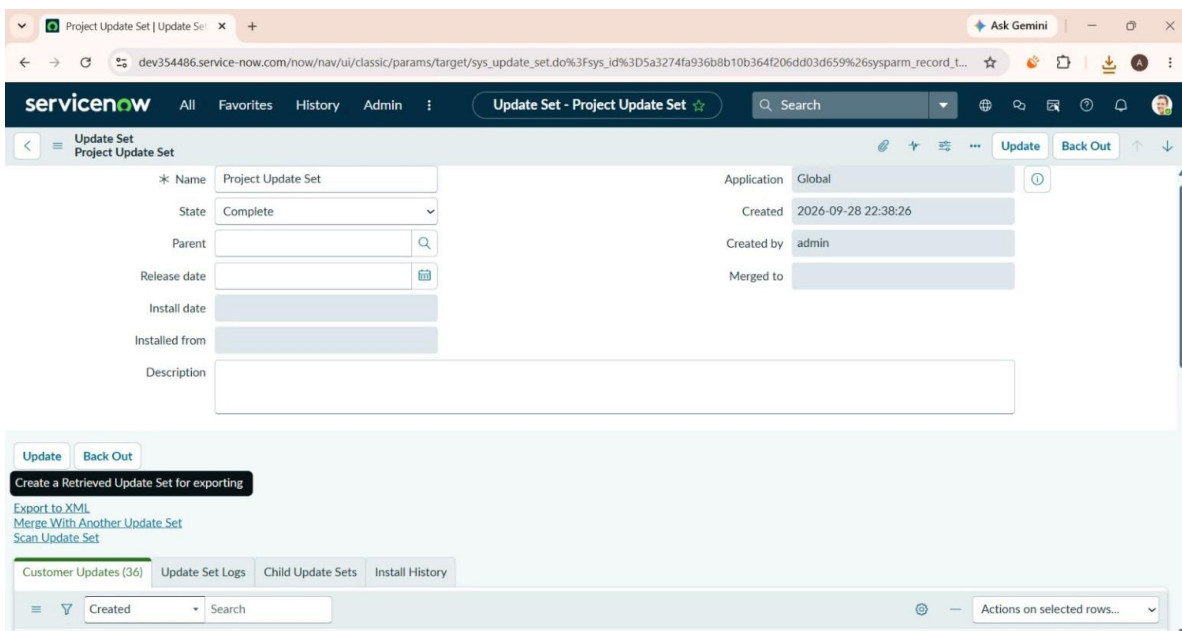


Figure 2. Project Update Set in Complete state.

4. Setup Instructions

Prerequisites:

- Authorized ServiceNow Developer Instance access.
- Required project/team permissions.
- Web browser with access to the ServiceNow instance.

Installation / Configuration Steps:

1. Open the authorized ServiceNow Developer Instance.
2. Verify the Incident Workflow table and form.
3. Verify Category and Subcategory values.
4. Verify that Subcategory is dependent on Category.
5. Open Flow Designer and verify the Auto Classify School IT Tickets flow.
6. Verify the trigger and classification branches.
7. Verify the Update Record actions.
8. Verify the Project Update Set and captured customer updates.

Credentials: The project documentation does not contain usernames or passwords. The team should use its authorized ServiceNow credentials.

5. Folder Structure

This project does not use a traditional source-code folder structure because it is implemented inside ServiceNow. The equivalent project structure is the ServiceNow configuration structure:

- Incident Workflow Table – stores project ticket records.
- Incident Workflow Form – provides the user interface.
- Category Field – stores the ticket category.
- Subcategory Field – stores the ticket subcategory.
- Dependent Field Configuration – controls Subcategory values based on Category.
- Flow Designer – contains the classification trigger, conditions and actions.
- Access Controls – provide controlled access to the custom table.
- Project Update Set – captures project configuration changes.

The screenshot shows the ServiceNow Form Designer interface for the 'Incident Workflow' form. The browser address bar indicates the URL: `dev354486.service-now.com/$ng_fd.do?sysparm_attributes=startTable='u_incident_workflow'&%2CstartView:'Default%20view'&sysparm_domain_restore=false&sysparm_stack=no`. The interface is titled 'Form Design' and shows the 'Incident Workflow [u_incident_workflow]' form. The left sidebar contains a 'Fields' section with a 'Filter' input and a list of fields: 'Created', 'Created by', 'Updated', 'Updated by', and 'Updates'. Below this is a 'Formatters' section with 'Activities (filtered)', 'Contextual Search Results', and 'Ratings'. The main area displays the form layout with fields: 'Number', 'Caller', 'Category', 'Subcategory', 'Short Description', 'Description', 'State', 'Assigned Group', and 'Assigned to'. Each field has a configuration icon (gear) and a delete icon (X).

Figure 3. Incident Workflow form and project field structure.

6. Running the Application

The ServiceNow project is executed through the ServiceNow platform rather than by running local npm commands.

1. Log in to the authorized ServiceNow Developer Instance.
2. Open the Incident Workflow form.
3. Create an Incident Workflow record.
4. Enter the relevant issue information.
5. Save/submit the ticket.
6. Allow the Flow Designer process to evaluate the configured conditions.
7. The matching condition determines the Category/Subcategory classification.
8. The Update Record action updates the Incident Workflow record.
9. Review the resulting ticket and confirmation notification where applicable.

Supported Classification Rules:

- Wi-Fi → Category: Network → Subcategory: Wi-Fi
- Projector → Category: Hardware → Subcategory: Projector
- Forgot Password → Category: Access → Subcategory: Forgot Password
- Slow Computer → Category: Performance → Subcategory: Slow Computer

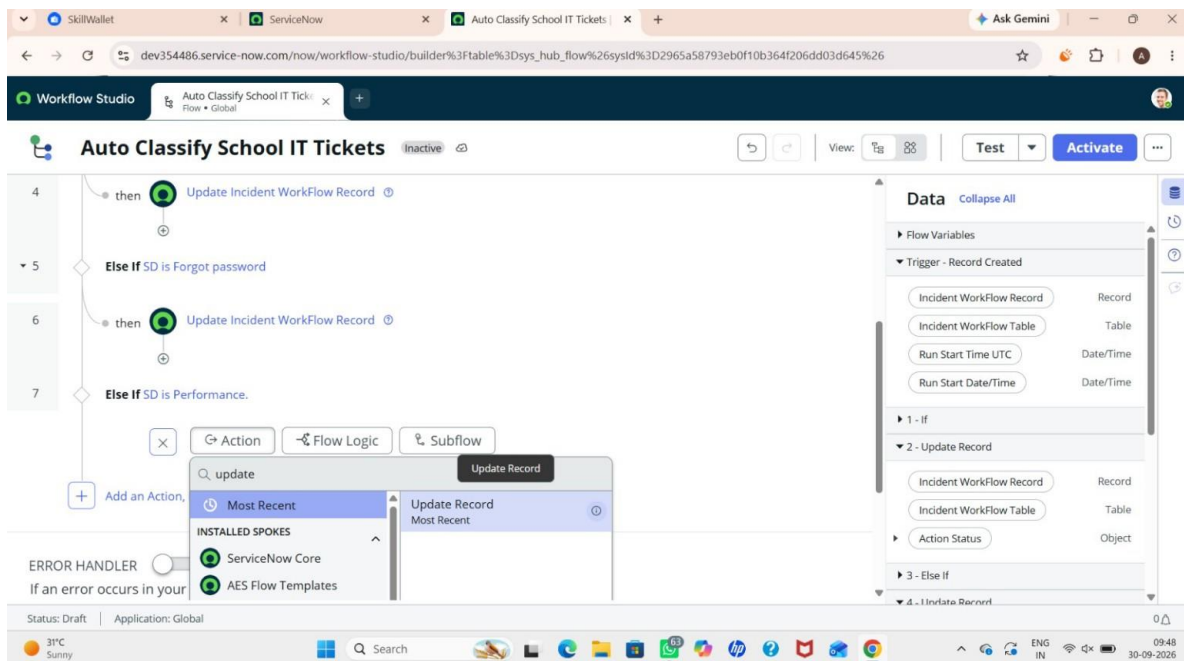


Figure 4. Flow Designer classification conditions and actions.

7. API Documentation

This project does not expose a separate backend API. The equivalent automation documentation is the Flow Designer configuration.

Flow Name: Auto Classify School IT Tickets

Trigger: Incident Workflow record creation requiring classification.

Classification Conditions:

- Wi-Fi → Network
- Projector → Hardware
- Forgot Password → Access
- Slow Computer → Performance

Action: Update Incident Workflow Record with the resulting classification.

Output: Updated ticket record and configured ticket confirmation notification.

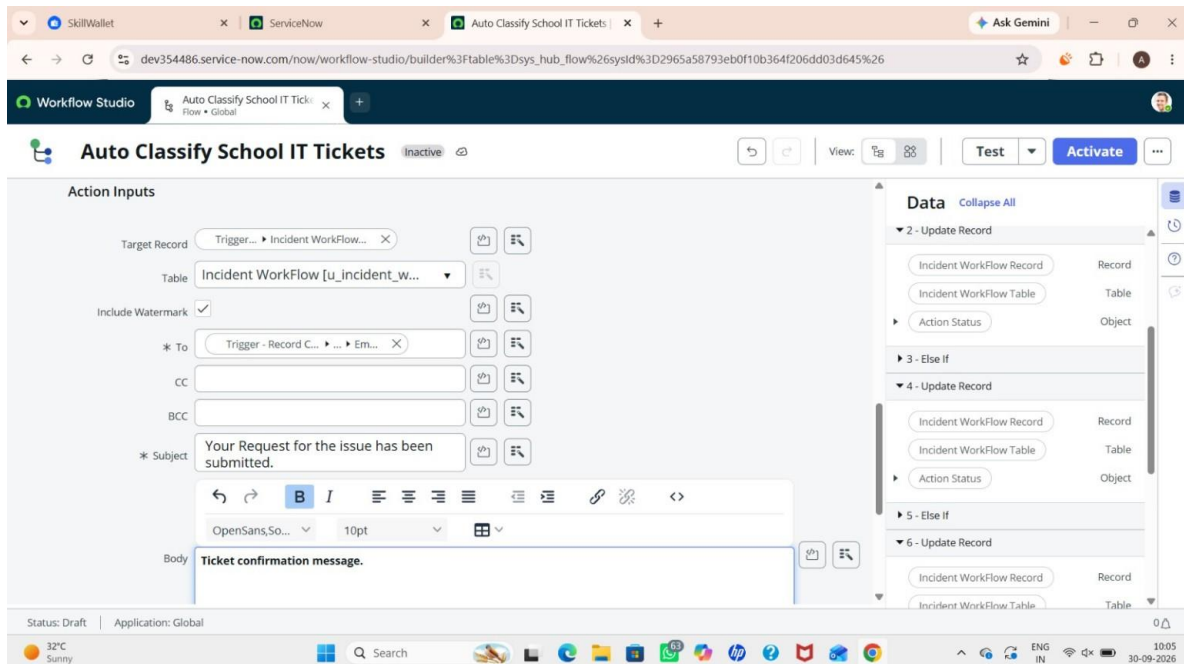


Figure 5. Update Record action in Flow Designer.

8. Authentication

Authentication is handled by the ServiceNow platform through the team's authorized account.

Authorization is controlled through the configured ServiceNow access-control mechanism and the project role for the custom Incident Workflow table.

Security Note: Passwords, tokens and other authentication secrets must not be included in the project report or screenshots.

9. User Interface

The primary user interface is the ServiceNow Incident Workflow form. It provides the fields required to create and review project tickets.

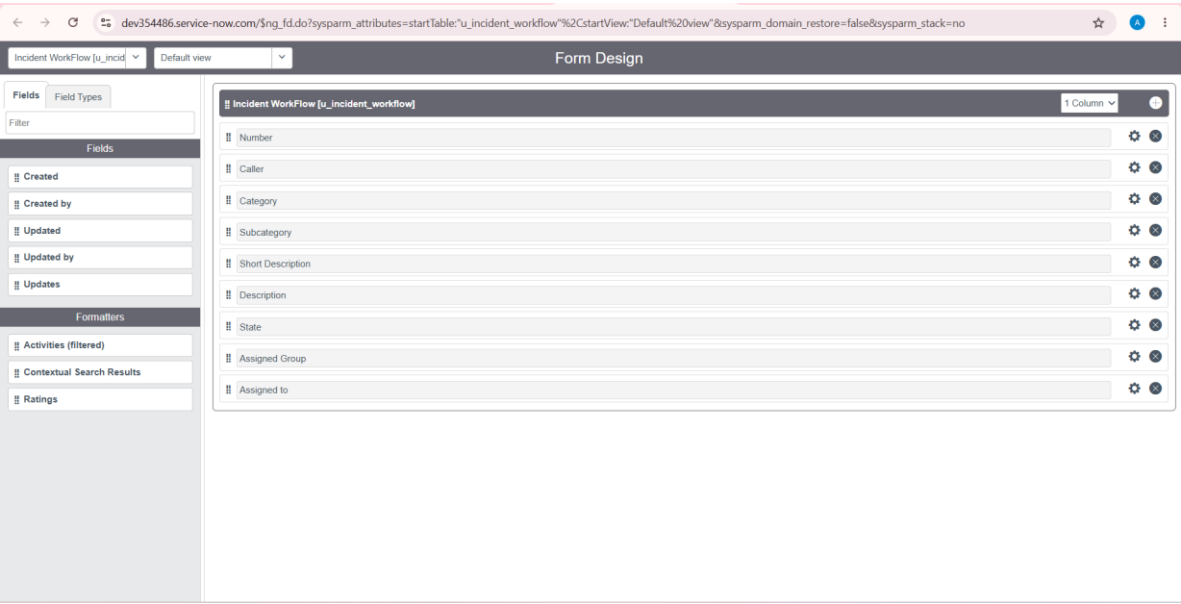


Figure 6. Incident Workflow user interface.

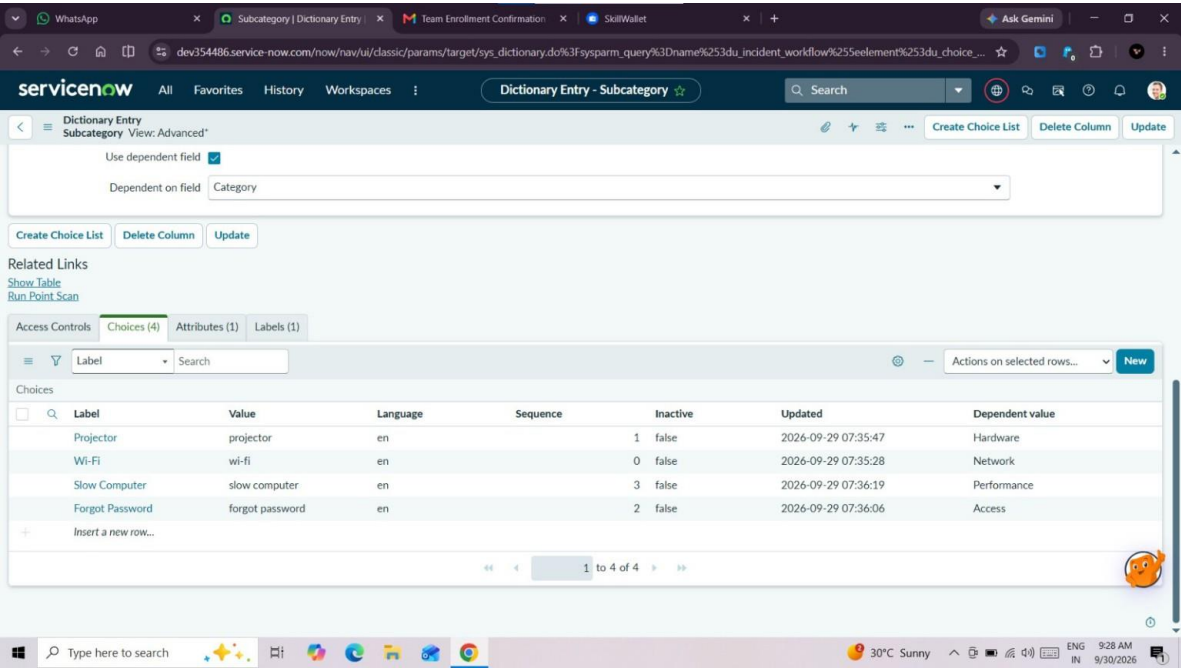


Figure 7. Category/Subcategory mapping configuration.

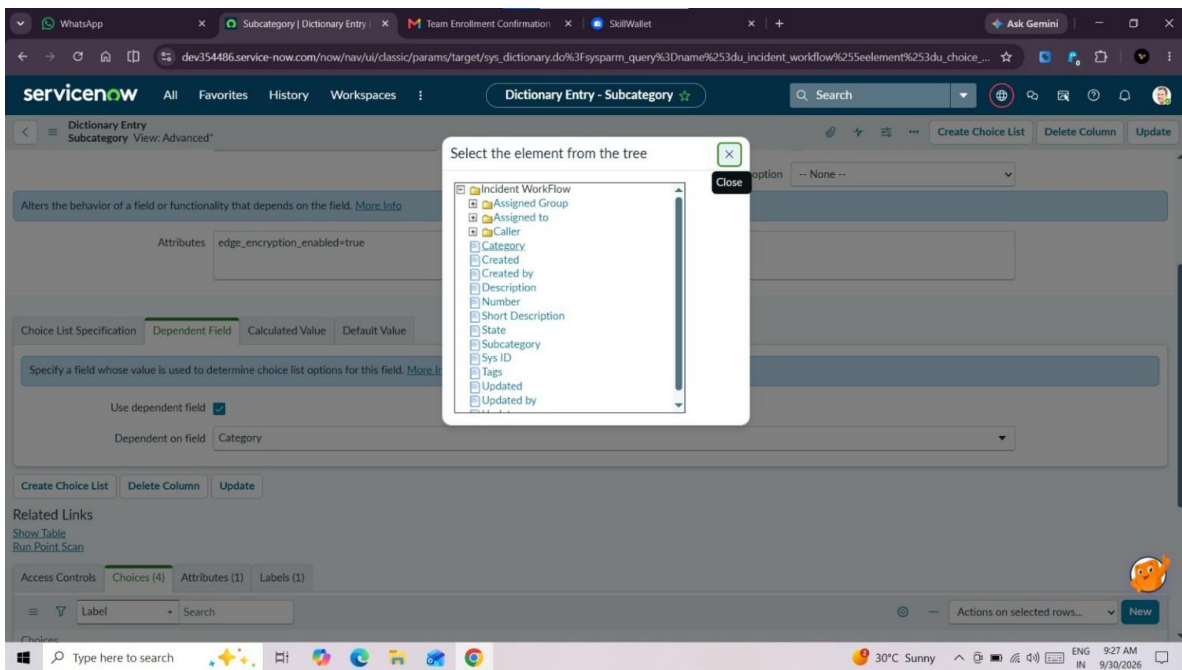


Figure 8. Subcategory dependency on Category.

10. Testing

Testing Strategy: Testing focuses on functional verification of ticket classification, dependent-field behavior, Flow Designer conditions, record update actions and notification output.

- FT-01 – Wi-Fi ticket classification: Expected Category = Network.
- FT-02 – Projector ticket classification: Expected Category = Hardware.
- FT-03 – Forgot Password classification: Expected Category = Access.
- FT-04 – Slow Computer classification: Expected Category = Performance.
- FT-05 – Dependent Subcategory validation: Subcategory values are controlled by Category.
- FT-06 – Blank Category condition: Configured flow evaluates the ticket for classification.

Performance Note: This is a rule-based Flow Designer automation, not a trained machine-learning model. Therefore, ML training accuracy, validation accuracy and confidence scores are not applicable. Unsupported response-time figures have not been invented.

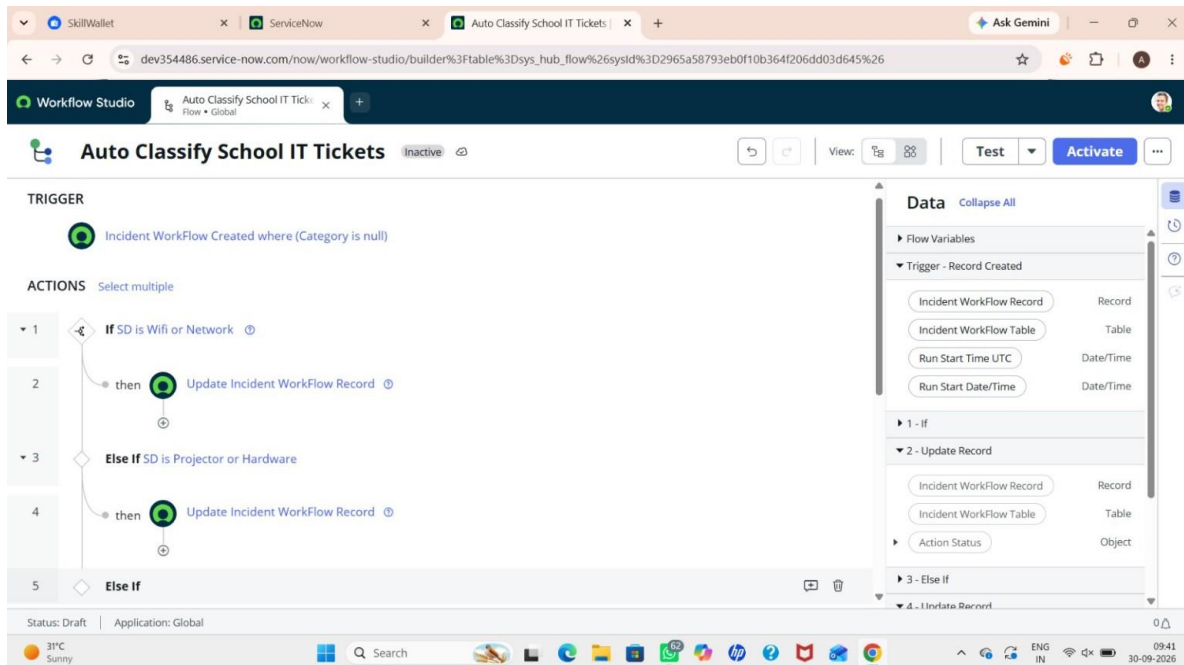


Figure 9. Flow Designer classification logic used as testing/configuration evidence.

11. Screenshots or Demo

Only screenshots that directly demonstrate the ServiceNow implementation or its output are included.

The screenshot shows the 'Form Design' interface for the 'Incident Workflow [u_incident_workflow]' form. The left sidebar contains a 'Fields' section with a list of fields: Created, Created by, Updated, Updated by, and Updates. Below this is a 'Formatters' section with 'Activities (filtered)', 'Contextual Search Results', and 'Ratings'. The main area displays the form layout with fields: Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned to. Each field has a configuration icon (gear) and a delete icon (X).

Figure 10. Incident Workflow form.

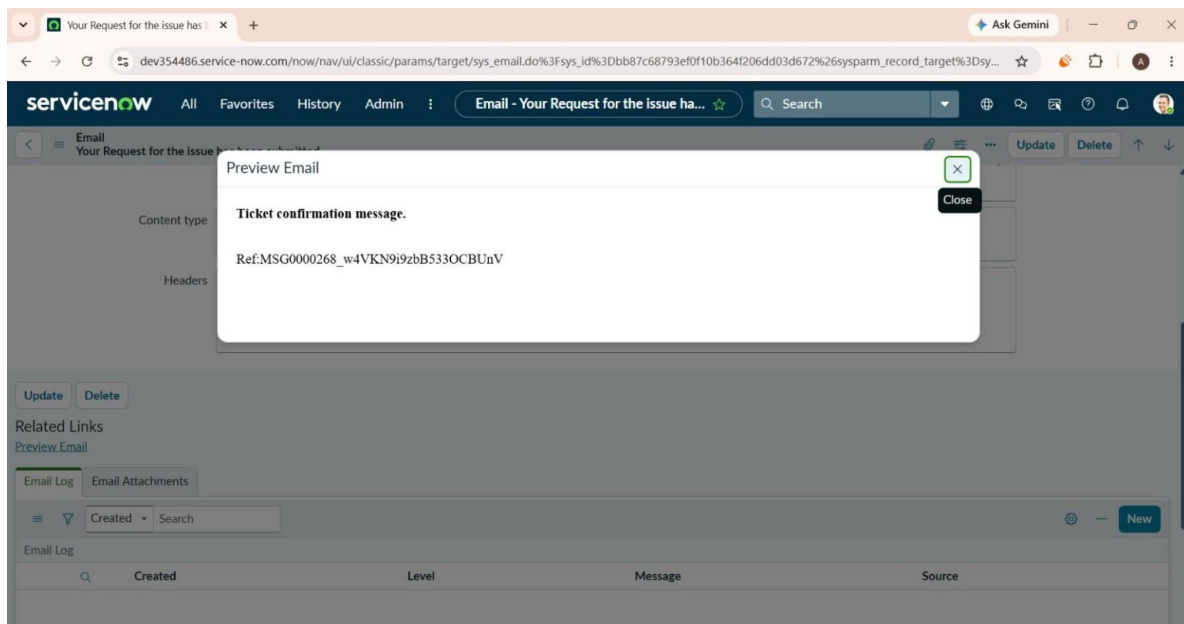


Figure 11. Ticket confirmation email preview. This demonstrates notification output.

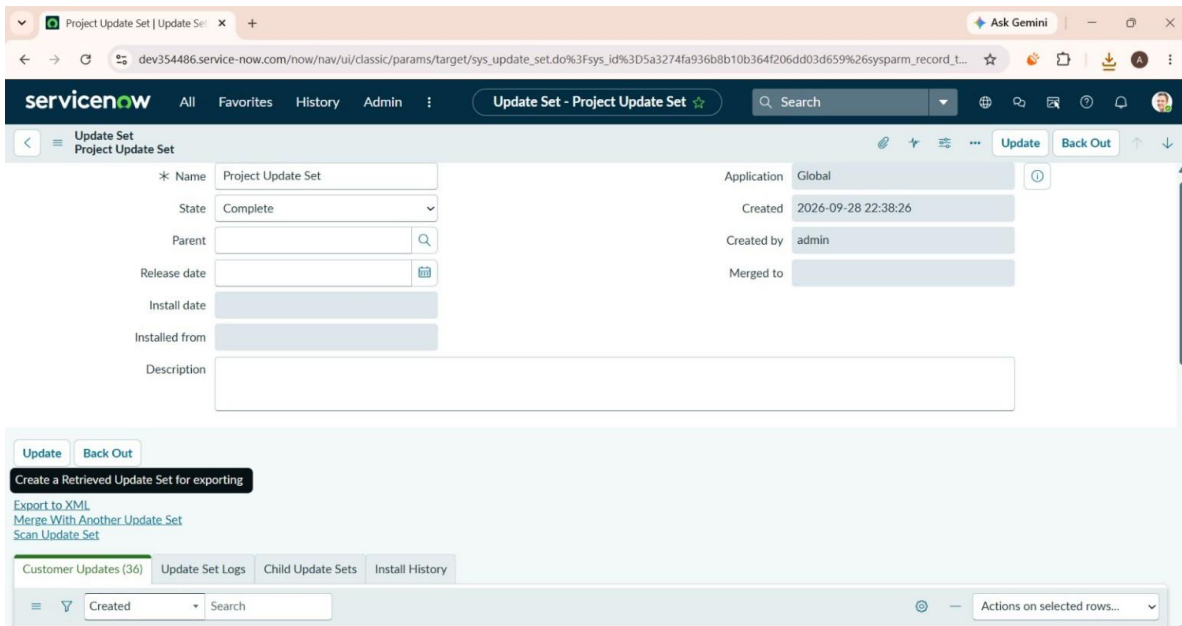


Figure 12. Project Update Set showing Complete state.

Demo Procedure:

1. Open the authorized ServiceNow Developer Instance.
2. Open the Incident Workflow form.
3. Create a supported issue ticket.
4. Submit/save the ticket.
5. Allow the configured flow to process the record.
6. Review the updated classification.
7. Review the confirmation notification where applicable.

12. Known Issues

- The current flow handles only the issue patterns explicitly configured.
- New issue types require additional conditions and testing.
- Unexpected or ambiguous issue descriptions may not match an existing rule.
- The current implementation is rule-based and does not use machine learning or Natural Language Understanding.
- Production-scale load benchmarking is not included in the available project evidence.
- Correct execution depends on ServiceNow configuration, permissions and flow activation.

13. Future Enhancements

- Add additional IT categories and subcategories.
- Use Natural Language Understanding or machine learning for free-text classification.
- Automatically assign tickets to appropriate support groups.
- Recommend ticket priority and impact.
- Add escalation and notification workflows.
- Create ServiceNow reporting dashboards for classification trends.
- Add automated regression testing for new classification rules.
- Measure production response time and throughput against defined operational targets.
- Add audit reporting for classification changes.